Interventional Radiology Management of Adult Liver Transplant Complications

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Liver transplant remains the definitive therapy for patients with end-stage liver disease. Outcomes have continued to improve, in part owing to interventions used to treat posttransplant complications involving the hepatic arteries, portal vein, hepatic veins or inferior vena cava (IVC), and biliary system. Significant hepatic artery stenosis can be treated with angioplasty or stent placement to prevent thrombosis and biliary ischemic complications. Hepatic arterioportal fistula and hepatic artery pseudoaneurysm are rare complications that can often be treated with endovascular means. Treatment of hepatic artery thrombosis can have mixed results. Portal vein stenosis can be treated with venoplasty or more commonly stent placement. The rarer portal vein thrombosis can also be treated with endovascular techniques. Hepatic venous outflow stenosis of the hepatic veins or IVC is amenable to venoplasty or stent placement. Complications of the bile ducts are the most encountered complication after liver transplant. When not amenable to endoscopic intervention, biliary stricture, bile leak, and ischemic cholangiopathy can be treated with percutaneous transhepatic cholangiography with biliary drainage and other interventions. New techniques have further improved care for these patients. Transsplenic portal vein recanalization has improved transplant candidacy for patients with chronic portal vein thrombosis. Spontaneous splenorenal shunt and splenic artery steal syndrome (nonocclusive hepatic artery hypoperfusion syndrome) remain complicated topics, and the role of endovascular embolization is developing. When patients have recurrence of cirrhosis after transplant, most commonly due to viral hepatitis, transjugular intrahepatic portosystemic shunt (TIPS) may be required to treat symptoms of portal hypertension.

Online supplemental material is available for this article.

Abbreviations: APF = arterioportal fistula, ERCP = endoscopic retrograde cholangiopancreatography, HAS = hepatic artery stenosis, HAT = hepatic artery thrombosis, IVC = inferior vena cava, PVS = portal vein stenosis, PVT = portal vein thrombosis, SMV = superior mesenteric vein, TIPS = transjugular intrahepatic portosystemic shunt

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Describe surgical techniques for liver transplant and anatomy after liver transplant.
- Discuss the role of interventional radiology in managing liver transplant complications, including indications, techniques, and expected outcomes.
- Understand new strategies and techniques that can be used after liver transplant.

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Introduction

Liver transplantation remains the definitive therapy for end-stage liver disease. In the United States, the number of liver transplants has increased by more than 40% in the past 10 years. Outcomes have also continued to improve, with 5-year survival rates in deceased-donor liver transplant exceeding 75% (1). Improvements in patient selection, surgical technique, pre- and posttransplant medical management, diagnostic imaging, and posttransplant interventions have all contributed to better outcomes.

Interventional radiology plays a critical role in management of liver transplant patients, providing minimally invasive treatment options. Complications after liver transplant can occur in the hepatic artery, portal vein, hepatic venous outflow, or biliary system. In this article, we discuss the surgical anatomy of liver transplant and review interventional radiology management of vascular and biliary complications after liver transplant. While abscess or fluid drainage

TEACHING POINTS

- Understanding the surgical anatomy is critical to recognizing the underlying disease in liver transplant complications. Reading the operative note is imperative, as there are individual and institutional differences in technique.
- If untreated, HAS can progress to hepatic artery thrombosis (HAT), which can lead to significant morbidity, graft loss, and need for retransplantation. Bile ducts in liver transplant patients are particularly prone to arterial ischemia, which can lead to biliary strictures or ischemic cholangiopathy.
- The most common complications after liver transplant involve the bile ducts and can result in significant morbidity and mortality.
- Percutaneous recanalization of a chronically occluded portal vein in conjunction with TIPS placement can be used to recanalize the portal vein and increase transplant candidacy.
- Also known as nonocclusive hepatic artery hypoperfusion syndrome, splenic artery steal syndrome is an underrecognized cause of liver dysfunction after liver transplant, as its manifestation mimics other causes of graft failure.

and transjugular biopsy are common procedures in this population, this review focuses on transplant-related interventions (Table 1). An update on transplant-related procedures is provided, such as portal vein recanalization before transplant, embolization of competing varices to increase portal flow, splenic artery steal syndrome, and transjugular intrahepatic portosystemic shunt (TIPS) after liver transplant.

Surgical Anatomy

Understanding the surgical anatomy is critical to recognizing the underlying disease in liver transplant complications. Reading the operative note is imperative, as there are individual and institutional differences in technique. Accordingly, review of US and cross-sectional imaging studies is important. The anastomoses that need to be considered are the hepatic artery, portal vein, hepatic veins and inferior vena cava (IVC), and bile ducts. Complications can occur at any of these sites.

Hepatic Artery

One of the more common hepatic artery surgical anastomoses is between the recipient proper hepatic artery and the donor common hepatic artery (Fig 1). Even with this configuration, there are multiple variations, such as an anastomosis between the recipient common hepatic artery and the donor common hepatic artery. Likewise, the gastroduodenal artery may remain open or be ligated, depending on surgeon preference.

There are numerous mesenteric arterial variants—such as replaced or accessory hepatic arteries—that need to be addressed before liver transplant. One relatively common variant is a donor with a replaced right hepatic artery (Fig 2). The replaced right hepatic artery can be anastomosed to the donor gastroduodenal artery on the back table. A common hepatic to common hepatic arterial anastomosis is then performed. Another variant is a donor replaced left hepatic artery arising from the left gastric artery. In this scenario, the splenic artery and branch arteries are ligated, and the donor celiac trunk is anastomosed to the recipient common hepatic artery (Fig 3).

In some settings, a bypass graft is needed, such as in retransplantation, in flow-limiting inflow dissection, or if a patient has a diseased celiac artery. Most commonly, the graft extends from the recipient infrarenal aorta to the donor common or proper hepatic artery. While polytetrafluoroethylene (PTFE) can be used, cadaveric iliac artery is favored (Fig 4). When a graft or conduit is used, two anastomoses are present. The proximal one is at the aortoconduit junction; the distal one is between the conduit and the common or proper hepatic artery.

Portal Vein

Fewer anatomic variants exist with the portal vein. Most commonly, the donor portal vein is anastomosed to the recipient portal vein (Fig 5). Patients with cirrhosis often develop an acute or chronic portal vein thrombus, which can progress to cavernous transformation. This can complicate the portal vein anastomosis, sometimes requiring portal vein thrombovenectomy at the time of transplant or a bypass conduit from the superior mesenteric vein (SMV) or portosplenic confluence to the donor portal vein. Cadaveric vein can be used (Fig 6), or an interposition graft can be used.

Hepatic Veins and IVC

Most commonly, the piggyback technique is used for the outflow veins. A common cloaca is made with the recipient hepatic veins to form one large vein. The donor liver along with the suprahepatic and infrahepatic IVC are prepared on the back table. The infrahepatic IVC is ligated. The donor suprahepatic IVC is anastomosed to the cloaca of the recipient (Fig 7).

Another option is a cavocavostomy, whereby the donor IVC (ligated suprahepatic and infrahepatic IVC) along with the donor liver are anastomosed to the recipient IVC by making an anterior slit, with the native hepatic veins ligated (Fig 8).

Bile Ducts

In patients with conventional anatomy, an anastomosis between the donor common bile duct and recipient common bile duct can be performed (choledocho-choledochostomy). In patients with a diseased common bile duct or undergoing retransplant or revision, a biliary-enteric anastomosis may

Table 1: Overview of Complications after Liver Transplant

Complications by Type	Onset after Transplant*	Treatment Options
Hepatic artery complications		
Hepatic artery stenosis (HAS) (4%–11%)†	Early	Surgical revascularization Endovascular intervention
	Late	Endovascular intervention Surgical revascularization (if not amenable to endovascular techniques)
Hepatic artery thrombosis (HAT) (5%–9%)	Early	Surgical revascularization Endovascular intervention Retransplantation
	Late	Observation
Hepatic arterioportal fistula (APF) (5%)	Early or late	Observation initially, particularly early as most resolve; if enlarging or reversed flow in first-order portal vein, endovascular intervention
Hepatic artery pseudoaneurysm (0.3%–2.6%)	Early > late	Extrahepatic pseudoaneurysm treated with endovascular intervention or surgical repair (if not amenable to endovascular intervention) Intrahepatic pseudoaneurysm treated with endovascular intervention or thrombin injection
Portal vein complications		
Portal vein stenosis (PVS) (1%–2%)	Late > early	Observation and surveillance with possible anticoagulation If symptomatic or high grade, endovascular intervention
Portal vein thrombosis (PVT) (1%–2.6%)	Perioperative (<72 hours)	Surgical revision Retransplantation
	Early	Endovascular intervention
	Late	If symptomatic, endovascular intervention favored If asymptomatic, observation or possible anticoagulation, depending on extent
Hepatic venous outflow complications (total 1%–6%)		
Hepatic vein outflow	Early	Surgical revision (rare) Endovascular intervention Observation
	Late	Endovascular intervention
IVC stenosis or thrombosis	Early	Surgical revision Endovascular intervention
	Late	Endovascular intervention
Biliary complications (total 5%–45%)		
Biliary stricture	Early	Surgical revision (rare) Endoscopic stent placement PTC with internal-external drainage
	Late (more common)	Endoscopic stent placement PTC with internal-external drainage
Bile leak	Early	Surgical revision Endoscopic stent placement PTC with internal-external drainage
	Late	Endoscopic stent placement PTC with internal-external drainage
Ischemic cholangiopathy	Late > early	Endoscopic stent placement PTC with internal-external drainage Retransplantation
Seroma, biloma, hematoma, abscess	Early	Postoperative collections may require percutaneous drainage or occasionally aspiration
Miscellaneous complications		
Neoplasm	Late	May require intra-arterial therapy or ablation for liver recurrence Recurrence in lung may be treated with ablation
Graft rejection	Late > early	Interventional radiology often plays a role in transjugular biopsy

Note.—IVC = inferior vena cava, > = more common than, PTC = percutaneous transhepatic cholangiography.
* Early = less than 30 days, late = greater than 30 days.
† Numbers in parentheses are the incidence.

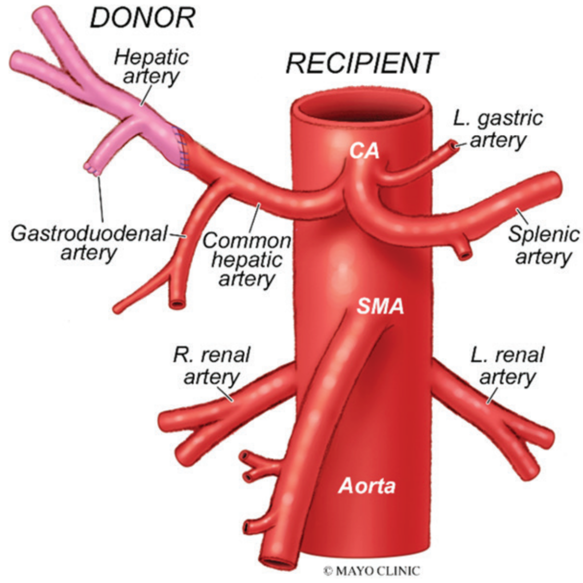


Figure 1. Conventional hepatic artery anastomosis. In this example, the recipient proper hepatic artery is anastomosed to the donor common hepatic artery. The recipient gastroduodenal artery may remain patent as shown or may be ligated. *CA* = celiac artery, *L.* = left, *R.* = right, *SMA* = superior mesenteric artery. (Used with permission from Mayo Foundation for Medical Education and Research, all rights reserved.)

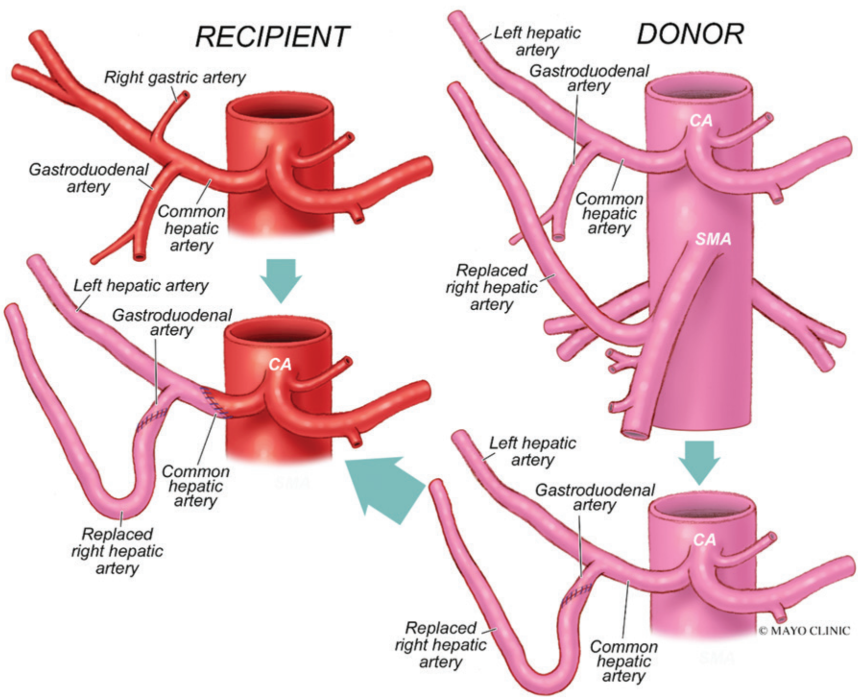


Figure 2. Hepatic artery anastomosis in a donor with a replaced right hepatic artery (rRHA). The donor has an rRHA arising from the superior mesenteric artery *(SMA)*. On the back table, the donor rRHA is anastomosed to the donor gastroduodenal artery. The donor common hepatic artery is anastomosed to the recipient common hepatic artery. *CA* = celiac artery. (Used with permission from Mayo Foundation for Medical Education and Research, all rights reserved.)

be necessary. In this scenario, the common bile duct or hepatic duct is anastomosed to a bowel loop, which is often a Roux-en-Y jejunal loop (Fig 9) or the duodenum.

Hepatic Artery Complications

Hepatic Artery Stenosis

In liver transplant vascular interventions, hepatic artery stenosis (HAS) is one of the more commonly encountered complications, occurring in 4%–11% of all liver transplant recipients (2). HAS is divided into early onset (<30 days after transplant) and late onset (>30 days after transplant). Early-onset HAS can be treated with surgical revision or endovascular options, recognizing that there is some risk of vessel injury or rupture in a fresh anastomosis. In general, if the stenosis is amenable, first-line treatment of late-onset HAS is endovascular intervention.

If untreated, HAS can progress to hepatic artery thrombosis (HAT), which can lead to significant morbidity, graft loss, and need for retransplantation. Bile ducts in liver transplant patients are particularly prone to arterial ischemia, which can lead to biliary strictures or ischemic cholangiopathy (3).

Most cases are diagnosed at surveillance US, demonstrating a resistive index of less than 0.5

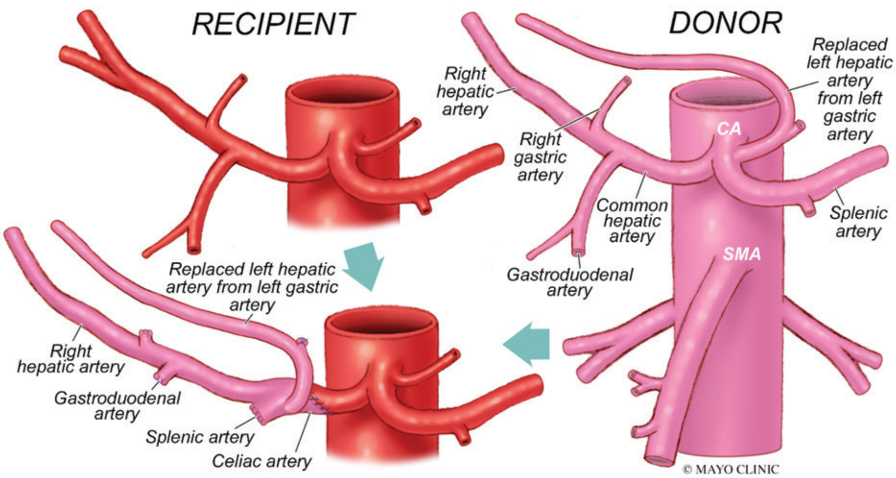


Figure 3. Hepatic artery anastomosis in a donor with a replaced left hepatic artery (rLHA). The donor has an rLHA arising from the left gastric artery (gastrohepatic trunk). The splenic artery, left gastric artery branch, gastroduodenal artery, and right gastric artery are ligated. The celiac artery *(CA)* from the donor is anastomosed to the recipient common hepatic artery. *SMA* = superior mesenteric artery. (Used with permission from Mayo Foundation for Medical Education and Research, all rights reserved.)

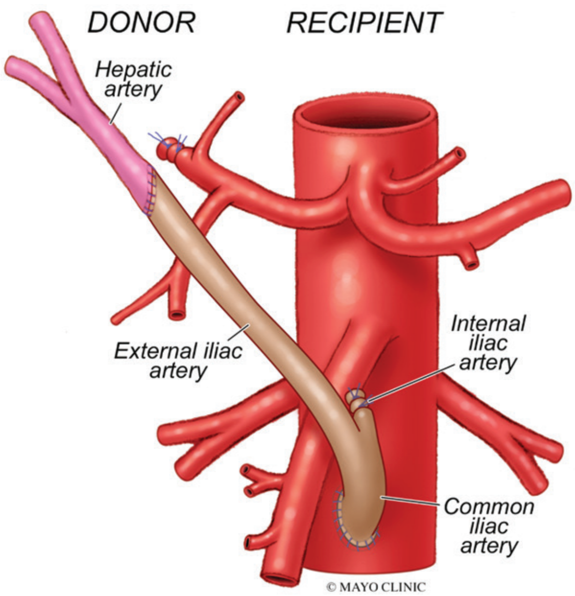


Figure 4. Hepatic artery anastomosis using a cadaveric iliac artery graft. In this example, a cadaveric iliac artery (with ligation of the internal iliac artery) is used as an infrarenal aorta graft to the donor common hepatic artery. When a graft is used, there are two anastomoses. (Used with permission from Mayo Foundation for Medical Education and Research, all rights reserved.)

(4) with elevated velocities at the anastomosis. However, clinical or laboratory abnormalities may prompt US evaluation. CT angiography is usually performed, which helps determine candidacy for intervention. CT angiography is also useful for procedure planning in deciding on a femoral or radial approach, obtaining measurements, and finding the optimal obliquity to assess stenoses in often tortuous vessels (Fig 10).

A guide catheter or sheath is placed in the celiac trunk or near the origin of the common hepatic artery. A 0.014-inch or 0.018-inch system is used to decrease vasospasm and safely cross the stenosis. Intra-arterial nitroglycerin can be administered via the sheath to also decrease vasospasm. Options for treatment include angioplasty and stent placement (5). Stent placement generally has lower rates of reintervention compared with angioplasty alone (6,7).

Our institution favors use of drug-eluting stents over bare metal stents for most cases of HAS, especially in small (<4 mm) arteries, to decrease the risk of restenosis related to intimal hyperplasia. One study showed a trend toward improved patency with drug-eluting stents; however, this difference did not reach statistical significance (8). Use of coronary stents in this vascular territory is considered off-label. Our institution's protocol is to give patients 75 mg of clopidogrel each day for 6 months and 81 mg of aspirin each day indefinitely. Complication rates for endovascular treatment are low (<5%), with dissection being the most commonly encountered (9).

A 2012 meta-analysis compared the effectiveness of angioplasty to that of stent placement in HAS and found both to be efficacious, with comparable patency and requirements for reintervention (5). A retrospective comparative single-institution study showed a higher 1-year patency rate with stent placement than with angioplasty (93.8% vs 73.5%) (10). Another large study showed a strong trend toward improved patency with stent placement versus with angioplasty (9). One-year primary patency rates for stents in HAS range from 68% to 90% (6,11), with the wide range likely due to differing study methodologies.

Hepatic Artery Thrombosis

HAT is a major complication that can occur days or years after liver transplant with an estimated prevalence of 4.8%–9% (12). Early-onset HAT is a serious complication with high morbidity and potential for graft loss without treatment. Treatment options include surgical revision, endovascular techniques, and retransplantation. Deciding on which to pursue depends on institution expertise

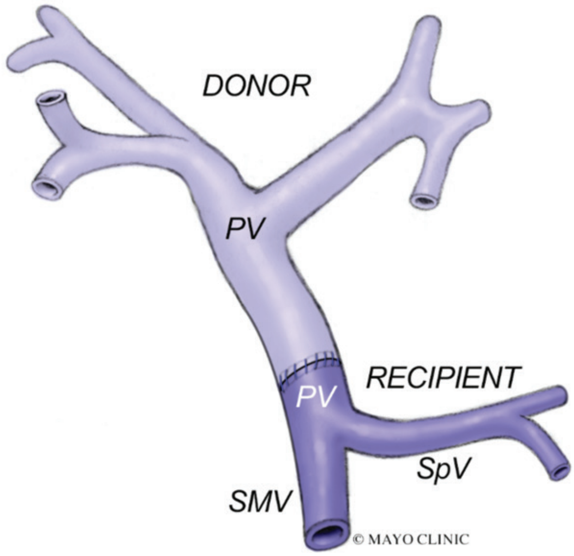


Figure 5. Portal vein anastomosis. When the portal vein *(PV)* is free of thrombus or disease, an end-to-end anastomosis may be performed between the donor portal vein and the recipient portal vein. *SMV* = superior mesenteric vein, *SpV* = splenic vein. (Used with permission from Mayo Foundation for Medical Education and Research, all rights reserved.)

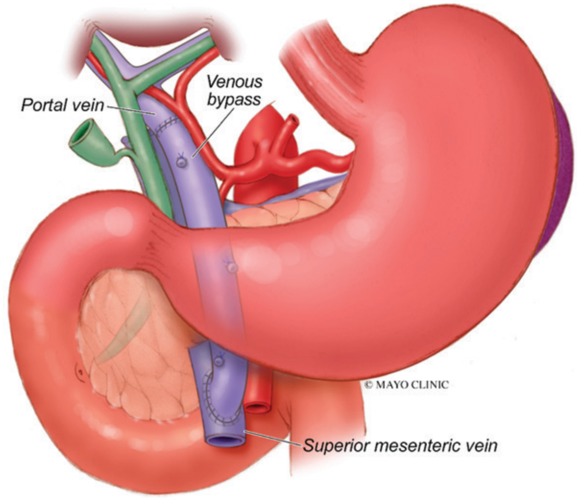


Figure 6. Bypass when an unsuitable portal vein is encountered. If the recipient portal vein is thrombosed or has undergone cavernous transformation, one technique is to perform a venous bypass using cadaveric vein. The bypass can extend from the SMV to the donor portal vein, as in this example. (Used with permission from Mayo Foundation for Medical Education and Research, all rights reserved.)

and requires a multidisciplinary discussion with transplant surgeons. At our institution, HAT within 7–10 days of transplant tends to be treated with surgical revision. Generally, an underlying stenosis is present, which could be at risk for rupture with angioplasty or stent placement.

Endovascular techniques include local thrombolysis (either on the table or with an overnight infusion), mechanical thrombectomy, angioplasty, and stent placement (13,14) (Fig 11). The results from endovascular treatments tend to be mixed, with much lower patency rates compared with those for nonthrombotic stenosis and with higher procedural complication rates. Late-onset HAT can have a more insidious manifestation and may be an incidental finding or detected with surveillance imaging. When the onset of HAS to HAT is gradual, collaterals can form from another hepatic artery branch or via extrahepatic supply, allowing perfusion to the intrahepatic branches of the liver (Fig E1). Intervention is generally not indicated in such cases.

Arterioportal Fistula

Arterioportal fistula (APF) is seen in up to 5.4% of patients after liver transplant (15). Most commonly, it represents intrahepatic APF, with extrahepatic APF being rare, particularly in the transplant setting. The most common causes are percutaneous procedures such as liver biopsy that are performed in the perioperative period. Most resolve spontaneously over time and can be monitored with surveillance US (Fig 12).

In about 0.2% of liver transplant patients, an APF can be hemodynamically significant, requiring treatment (16). Indications for intervention include a symptomatic APF causing bleeding, portal hypertensive sequelae, or graft dysfunction. Other indications include reversal of flow in the portal vein (first order or more central) or continued enlargement over time.

APF is amenable to endovascular treatment with coil or plug embolization (Fig 13). The technical success rate is high (63%–77%), with a complication rate of 12%–15% (15). The transplant hepatic artery can be challenging to navigate, depending on the anatomy. The fistula or communication may be difficult to identify, requiring embolization of the feeding artery and its vascular territory. Some fistulas may have multiple feeding arteries. All of these factors contribute to the complexity of treating APF, whereby repeat embolization is required in about 25% of cases (15).

Hepatic Artery Pseudoaneurysm

Hepatic artery pseudoaneurysm (HAPA) has an estimated prevalence of 0.3%–2.6% in liver transplant recipients, with an associated mortality rate approaching 75% (17). The manifestation of HAPA is variable, ranging from asymptomatic to life-threatening hemorrhage. Extrahepatic HAPA most commonly occurs at the anastomosis, with most cases diagnosed within 1 month of transplant (18). Risk factors include bacteremia, bile leak, abdominal infection, and iatrogenic interventions.

Endovascular options for repair include embolization using coils or liquid embolics and placing a

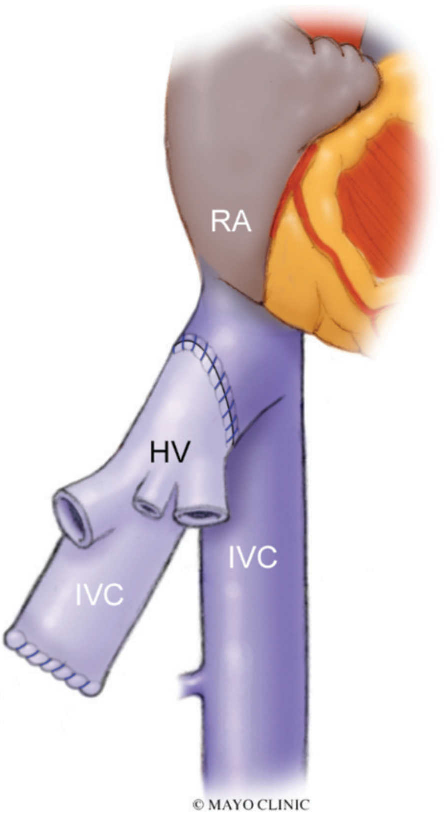


Figure 7. Hepatic outflow vein anatomy: piggyback technique. A cloaca is made from the hepatic veins *(HV)* of the recipient, making one large vein suitable for the anastomosis. The donor liver along with the suprahepatic IVC and infrahepatic IVC are brought to the surgical field. The infrahepatic IVC is ligated, leaving an often-visualized stump. The suprahepatic vein is anastomosed to the newly formed cloaca. *RA* = right atrium. (Used with permission from Mayo Foundation for Medical Education and Research, all rights reserved.)

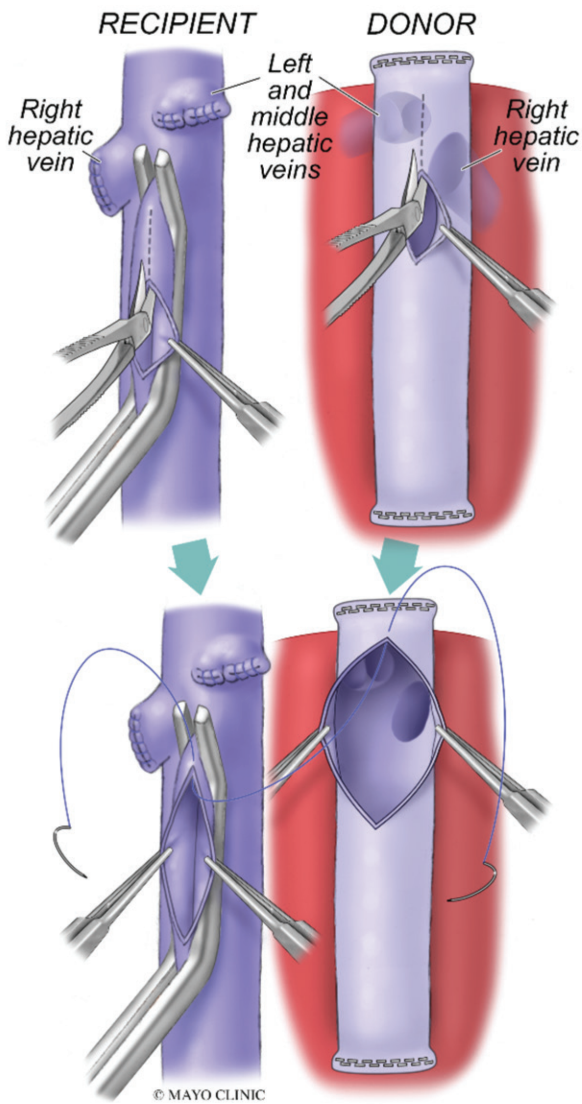


Figure 8. Cavocavostomy. The donor liver along with the suprahepatic and infrahepatic IVC are brought into the surgical field. The recipient hepatic veins have been ligated after removal of the native liver. The donor IVC is anastomosed to the recipient IVC as a cavocavostomy is performed. (Used with permission from Mayo Foundation for Medical Education and Research, all rights reserved.)

stent-graft across the aneurysm neck (19) (Fig 14). If embolization is undertaken, the goal should be to fill the aneurysm sac and preserve arterial flow to the graft. Surgical revision is generally reserved for cases not amenable to endovascular therapy.

Intrahepatic HAPA can occur owing to iatrogenic injury such as liver biopsy. This can be treated with transarterial embolization using coils or liquid embolic agents. Another option is US-guided percutaneous thrombin injection of the pseudoaneurysm.

Portal Vein Complications

Portal Vein Stenosis

Portal vein stenosis (PVS) is an uncommon entity in liver transplant patients, occurring in less than 1%–2% of adult whole-liver grafts (20), with a higher prevalence in pediatric living donor grafts (approaching 8% in one study) (21). PVS typically arises at the portal vein anastomosis and is identified on average more than 1 year after liver transplant secondary to scarring or fibrosis and neointimal hyperplasia. Severe PVS may lead to portal vein thrombosis (PVT), abnormal liver function test results, or sequelae of portal hypertension, such as variceal bleeding or ascites (20,22).

However, the majority of PVS cases encountered are asymptomatic and discovered at surveillance US. These patients can be followed up with surveillance US, with some consideration given to anticoagulation. Treatment is undertaken in symptomatic patients or those with a high-grade stenosis thought to be at risk for progression to PVT.

Endovascular intervention with angioplasty or stent placement represents the mainstay of PVS treatment. Several studies have demonstrated a

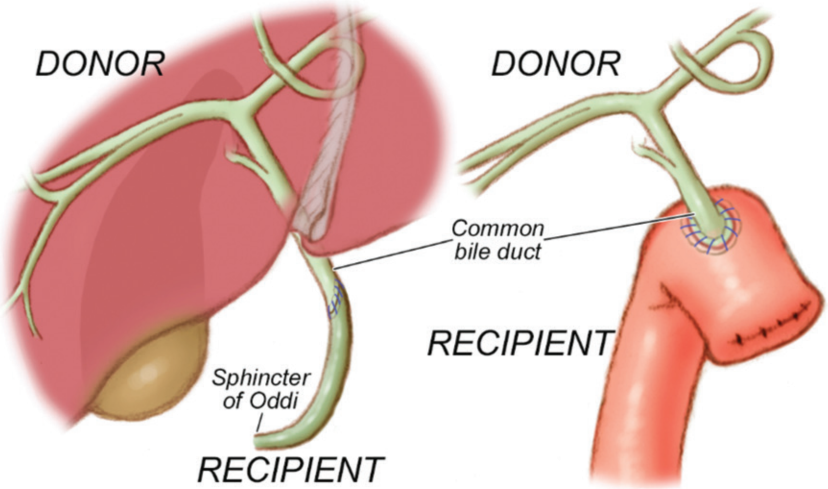


Figure 9. Biliary surgical anatomy. With conventional anatomy, the donor common bile duct is anastomosed to the recipient common bile duct (choledocho-choledochostomy) (left). If the recipient common bile duct is not amenable to surgical anastomosis, a biliary-enteric anastomosis may be required. As shown on the right, the common bile duct is anastomosed to a Roux-en-Y jejunal loop in an end-to-side fashion. (Used with permission from Mayo Foundation for Medical Education and Research, all rights reserved.)

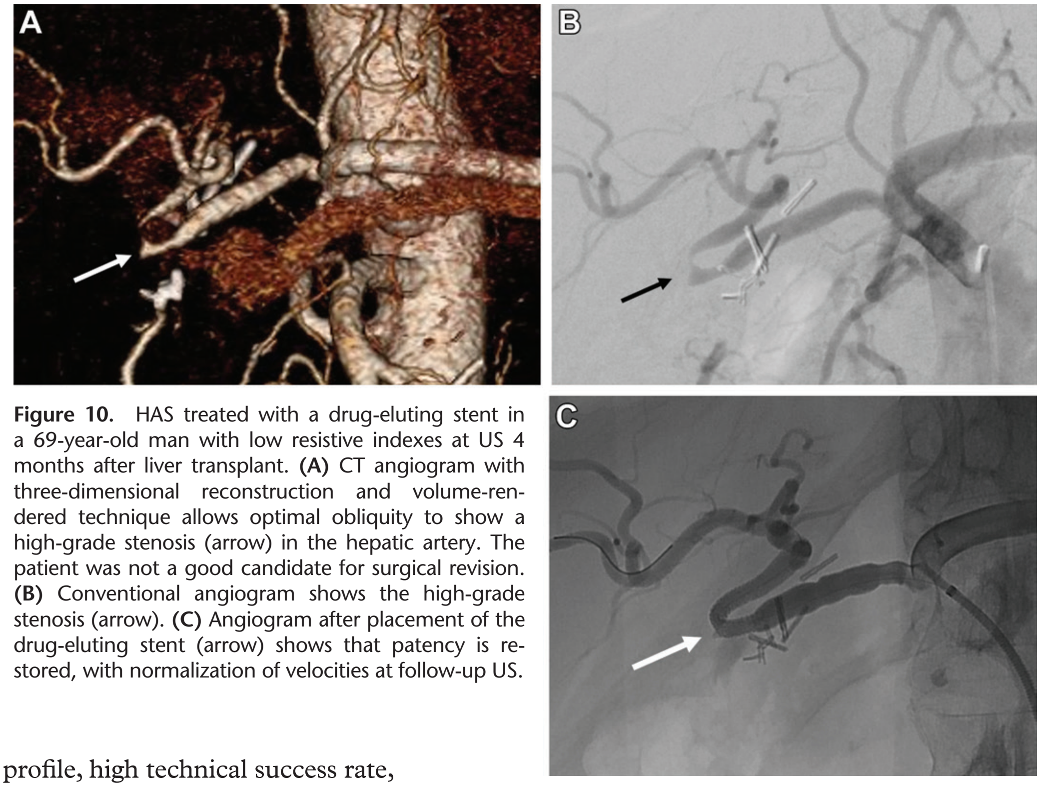


Figure 10. HAS treated with a drug-eluting stent in a 69-year-old man with low resistive indexes at US 4 months after liver transplant. **(A)** CT angiogram with three-dimensional reconstruction and volume-rendered technique allows optimal obliquity to show a high-grade stenosis (arrow) in the hepatic artery. The patient was not a good candidate for surgical revision. **(B)** Conventional angiogram shows the high-grade stenosis (arrow). **(C)** Angiogram after placement of the drug-eluting stent (arrow) shows that patency is restored, with normalization of velocities at follow-up US.

favorable safety profile, high technical success rate, and good long-term patency, specifically in the setting of adult posttransplant PVS (23,24). A retrospective study of posttransplant PVS patients who underwent transhepatic portal venoplasty followed by stent placement showed a technical success rate of 100% (25), with mean patency of 33 months. A more recent study compared angioplasty to stent placement in PVS patients and found that the 5-year patency rate for angioplasty alone was 82% but 100% in the stent group (26).

PVS intervention is typically performed via a transhepatic percutaneous intercostal approach; however, transsplenic access or a transjugular approach can be used (20). Access is obtained into a portal vein branch using US guidance. Initially, a 5-F sheath is placed. A wire and catheter are advanced past the stenosis into the SMV. Portal venography and pressure measurements are performed in the superior mesenteric vein and portal vein distal to the stenosis. In general, a pressure gradient above 4 mm Hg is considered hemodynamically significant; however, reported cutoffs range from 3 to 8 mm Hg (20,21).

Venoplasty or stent placement can be performed. Given the improved patency of stents and the risk of bleeding with repeated transhepatic interventions, our institution favors primary stent placement for PVS. Certainly, this depends on the anatomy, the result after the initial angioplasty, and retransplant candidacy.

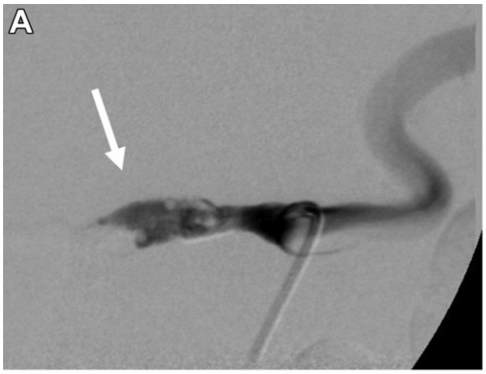
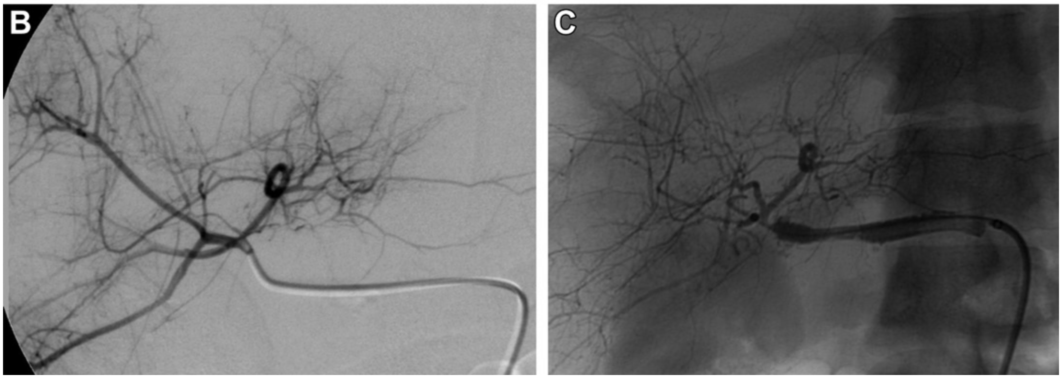


Figure 11. HAT in a 37-year-old man with undetectable flow at US surveillance 6 months after liver transplant. **(A)** Hepatic angiogram shows complete thrombosis (arrow) of the hepatic artery. After intra-arterial administration of 6 mg of tissue plasminogen activator (tPA) and heparin directly into the thrombus, the thrombus was crossed. **(B)** Angiogram shows patent intrahepatic artery branches. **(C)** The thrombosis was due to tandem lesions (not shown), which were ultimately treated with stent placement, with restoration of flow. The stent rethrombosed 1 month later (not shown).



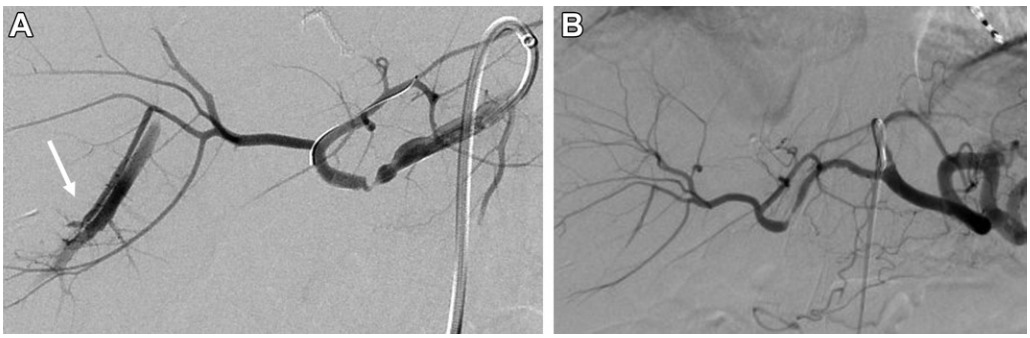


Figure 12. Spontaneous resolution of APF. **(A)** Hepatic angiogram obtained owing to HAS 4 months after liver transplant shows an APF (arrow). The incidental stenosis was treated with a bare-metal stent. **(B)** Angiogram obtained for in-stent restenosis 1 year later shows resolution of the APF. The restenosis was treated with a drug-eluting stent (not shown).

A self-expanding or balloon-expandable bare-metal or covered stent can be used (Fig 15).

Portal Vein Thrombosis

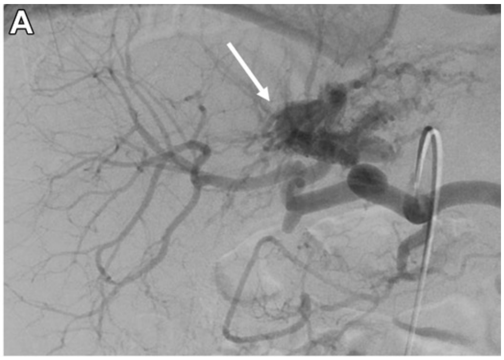
PVT is a rare complication after liver transplant, with variable prevalence depending on graft type and patient age. The prevalence of PVT in adult whole-liver grafts is estimated between 1.0% and 2.6% (27,28). The clinical manifestation includes symptoms of portal hypertension, such as ascites, gastrointestinal hemorrhage, graft failure, and abnormal liver function test results.

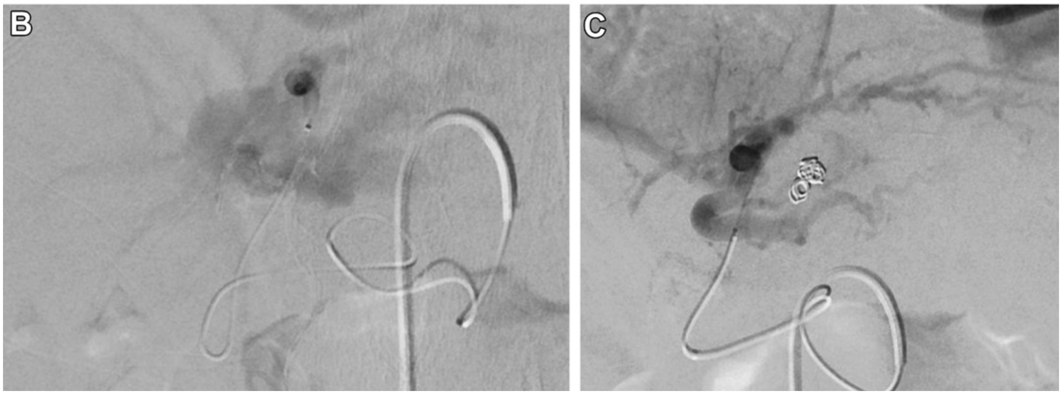
PVT can occur in the perioperative period (<72 hours from transplant), early (<30 days), or late (>30 days). In distinction from PVS, PVT more typically occurs in the early period. Perioperative PVT carries a poor prognosis, with decreased patient survival and a high rate of graft failure—up to 75% of cases (20). These cases are treated with surgical revision.

Early PVT can also have a poor prognosis; prompt detection and intervention are vital to reduce graft loss and patient mortality. In these patients, endovascular treatment can be undertaken. Options include a transhepatic, transjugular, or transplenic approach to the portal vein. A thrombolysis catheter can be placed in the thrombosed portal vein with an overnight infusion. Alternatively, thrombolysis and thrombectomy can be performed, all in one setting. Usually, an underlying stenosis is present, requiring stent placement (Fig 16).

A hybrid approach can also be used, with surgical exposure to a mesenteric vein, whereby an intervention can be performed from the mesenteric vein

Figure 13. Arterioportal fistula (APF) requiring treatment in a 55-year-old man after liver transplant with a known left APF followed with US surveillance. Over 1.5 years, hepatofugal flow developed in the left portal vein. Additionally, liver biopsy showed evidence of vascular compromise. **(A)** Angiogram shows a large APF (arrow) in the left liver with portal vein filling. **(B)** Angiogram obtained with a microcatheter in a left hepatic artery branch proximal to the fistula shows portal vein filling. **(C)** Angiogram obtained after coil embolization shows that no portal flow is present.





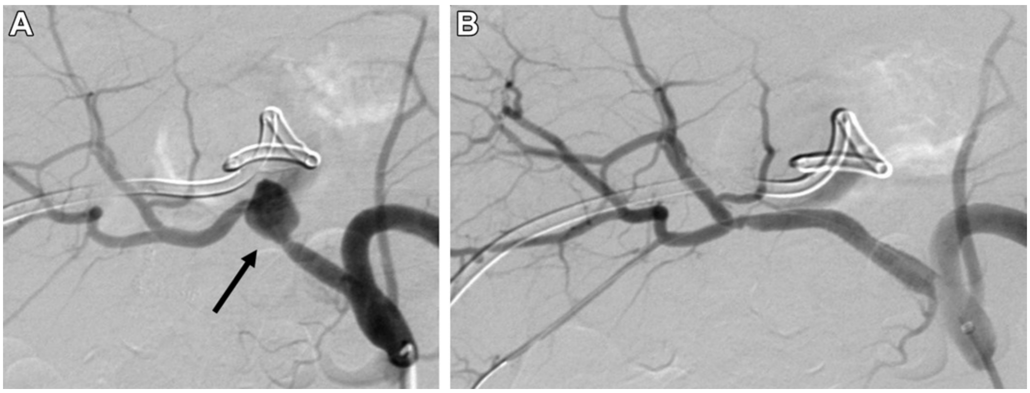


Figure 14. Extrahepatic anastomotic HAPA in a 55-year-old man after liver transplant 2 years earlier. **(A)** Hepatic angiogram shows a 1.5-cm anastomotic pseudoaneurysm (arrow). **(B)** Angiogram after placement of a stent-graft shows exclusion of the aneurysm with preservation of flow to the graft.

(29). Data are limited on endovascular treatment of PVT, with several adult case reports in the literature; the technical success rate ranges from 55% to 70% (20), with little data on long-term efficacy.

Late-onset PVT is rarer but may be treated with endovascular means. However, in asymptomatic patients, observation with anticoagulation is an option.

Hepatic Venous Outflow Obstruction

Hepatic venous outflow obstruction (HVOO) after liver transplant can be at the level of the hepatic veins or IVC. The prevalence of HVOO in orthotopic liver transplants is between 1% and 6% (30). The most common symptom of HVOO is refractory ascites, with others being allograft dysfunction (elevated serum enzyme levels), lower extremity edema, pleural effusion, abdominal pain, intestinal congestion, and splenomegaly (31). In advanced cases, fulminant graft dysfunction can occur, with a mortality rate as high as 24% (31,32). Loss of phasicity in the hepatic vein or IVC at Doppler US can suggest a stenosis. Cross-sectional imaging can be useful to confirm these findings.

Early complications will often be related to surgical technique, such as a tight anastomosis or twisted allograft causing vascular kinking. Treatment options include endovascular intervention, surgical revision, and retransplantation. Endovascular treatment may have limitations if the stenosis was due to kinking or torsion. Accordingly,

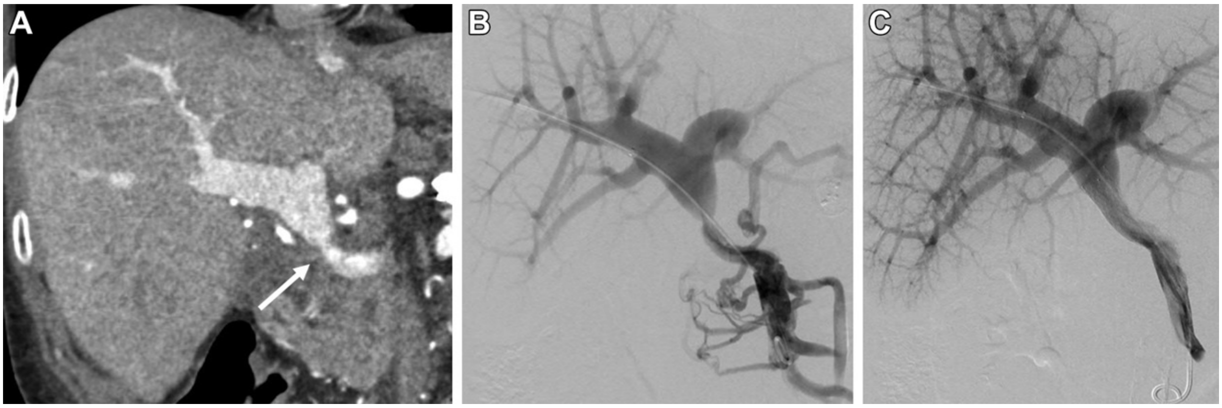


Figure 15. PVS in a 64-year-old woman 4 months after liver transplant. **(A)** Coronal contrast-enhanced CT image shows high-grade PVS (arrow) at the anastomosis, with velocities of more than 200 cm/sec at US. Transhepatic portal vein access was obtained, and a pigtail catheter was advanced into the SMV. **(B)** Venogram shows PVS with prominent collateralization and a 6 mm Hg pressure gradient across the stenosis. **(C)** Venogram after portal vein angioplasty and stent placement shows resolution of the pressure gradient and collaterals.

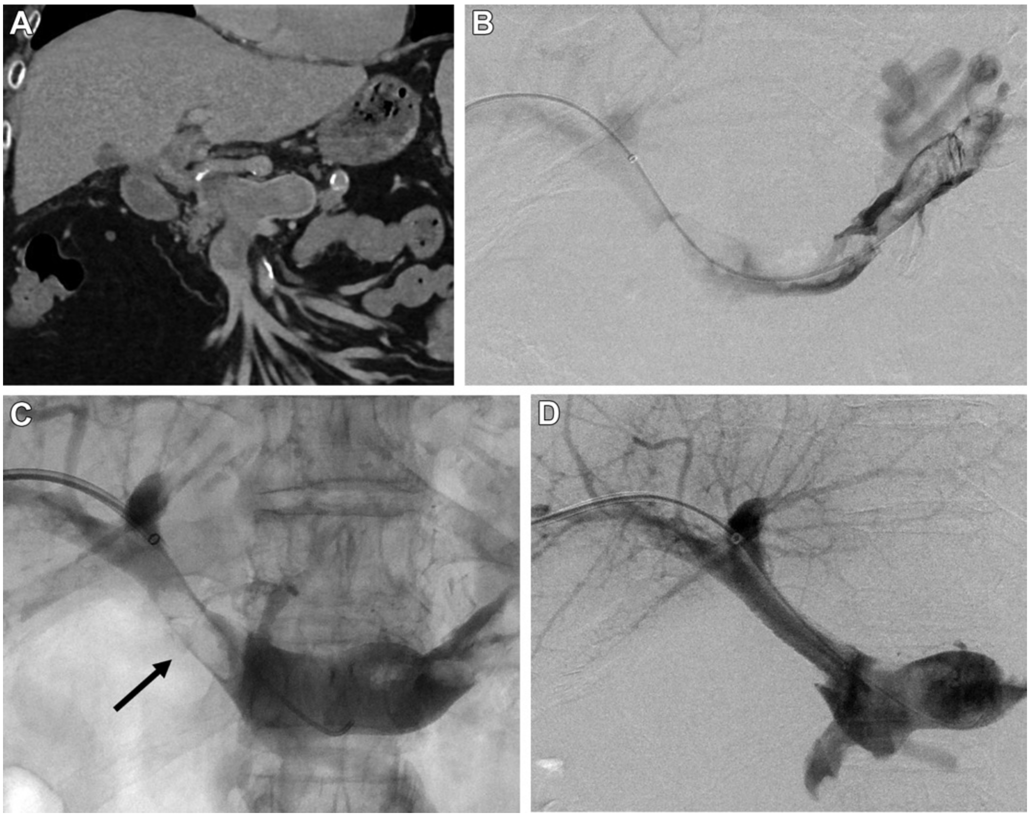


Figure 16. PVT in a 66-year-old woman with nausea and vomiting 6 months after liver transplant. **(A)** Coronal CT image shows PVT and splenic vein thrombosis, which extend into the SMV. **(B)** Venogram obtained with transhepatic portal vein access and catheter placement in the splenic vein shows extensive thrombus. A US-assisted thrombolysis catheter was placed for overnight infusion. On postprocedure day 1, additional thrombectomy was performed using a rheolytic mechanical thrombectomy device. **(C)** Venogram shows a filling defect (arrow), which persisted at the portal vein anastomosis despite continued thrombectomy and angioplasty. **(D)** Portogram after stent placement shows excellent flow.

there is a theoretical risk of vessel injury with a fresh anastomosis. Usually, late complications arise owing to neointimal hyperplasia, fibrosis due to inflammation, or venous compression due to graft maturation. Late HVOO is generally treated with endovascular techniques.

Hepatic Vein Stenosis

In hepatic vein stenosis, endovascular venoplasty with or without stent placement is the first-line treatment (31). Transjugular access is the preferred approach for endovascular treatment. In instances where the hepatic vein is unable to be selected from a jugular approach (often due to a high-grade stenosis), a femoral approach can be attempted. A percutaneous transhepatic approach to the hepatic veins can be a useful alternative (Fig E2).

The pressure gradient between the hepatic vein and right atrium is measured. The definition of a significant gradient is controversial, with a threshold between 3 and 10 mm Hg (33). The goal after intervention is reduction and resolution of a significant gradient. With venoplasty alone for hepatic vein interventions, recurrence is seen in up to 55% of patients, with potential need for serial dilation (31).

Improved patency of up to 92% may be achieved with use of stents during primary treatment. This is a challenging area for stent placement owing to proximity of the right atrium, cardiac motion, respiratory motion, and need for proper obliquity to identify the stenosis. Stent placement is safe, with the complication of stent migration being rare (34). Either a self-expanding stent or a balloon-expandable bare-metal stent may be used.

IVC Stenosis

The frequency of IVC stenosis has likely decreased with use of the piggyback technique, with an estimated prevalence of 0.8% (30). In IVC stenosis, recurrence with angioplasty alone is seen in up to 50% of patients (31), requiring serial dilation. Caval stent placement has a high success rate, with patency rates of up to 93% at 5 years, and should be considered for a durable outcome (32).

The Gianturco Z-stent (Cook Medical) has wide interstices and theoretically will not obstruct the hepatic venous outflow. The Wallstent (Boston Scientific) can be used as well (Fig E2). Stent migration and IVC rupture are potential serious complications of endovascular IVC venoplasty and stent placement; however, both are rare (31).

IVC Thrombosis

Thrombosis of the IVC is an uncommon complication after liver transplantation that can have high morbidity. When IVC thrombosis is diagnosed, treatment options range from conservative management to retransplantation. Conservative treatment with anticoagulation is generally reserved for nonocclusive thrombus.

Catheter-directed thrombectomy or thrombolysis is the most common treatment. Often, an underlying IVC stenosis is present, which also needs to be treated (Fig 17). Surgical thrombectomy followed by stent placement can also be an option for treatment. Percutaneous angioplasty and stent placement have a better overall outcome than surgical treatment (35).

Biliary Complications

The most common complications after liver transplant involve the bile ducts and can result in significant morbidity and mortality. These complications include biliary stricture (anastomotic and nonanastomotic), bile leak, and ischemic cholangiopathy. Other less common biliary-related complications include choledocholithiasis and hemobilia.

There is a wide range of reported total prevalence of biliary complications after liver transplant, between 5% and 45% (36). This variability is affected by the graft type: living versus deceased donation. Donation after cardiac death (DCD) has a higher rate of complications than donation after brain death (DBD), with living donation having the highest rate (37). Other factors include number of biliary anastomoses, ischemia time, and surgical technique.

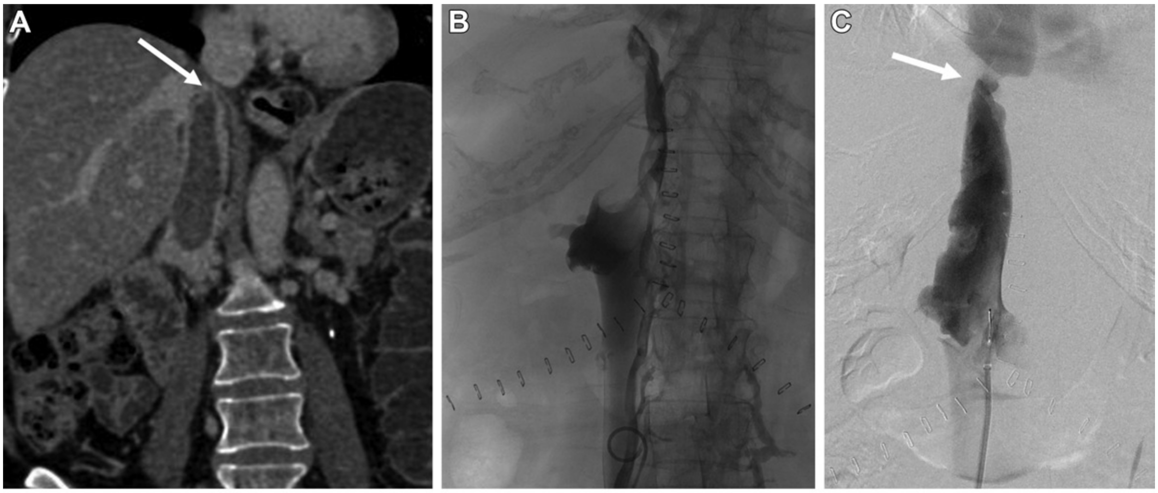
The type of biliary anastomosis dictates options for intervention, as choledocho-choledochal anastomoses are more amenable to endoscopic management with endoscopic retrograde cholangiopancreatography (ERCP) than are hepaticojejunostomies, which make biliary cannulation less straightforward. Therapeutic options are discussed in the context of each pathologic condition detailed in this section.

Biliary Stricture

Biliary strictures are seen in 5%–15% of patients after deceased donor liver transplant and 28%–32% of patients undergoing living donor transplant (36). Strictures can occur weeks to years after the transplant, with mean interval manifestation of 5–8 months (37). Presenting signs and symptoms include jaundice, hyperbilirubinemia, and cholangitis.

Strictures are divided into two types: anastomotic and nonanastomotic. Anastomotic strictures are more common and may be due to surgical technique, duct size discrepancy between donor and recipient, ischemia, prior infection, or fibrosis (38). They can occur at the choledocho-choledochal anastomosis or the biliary-enteric anastomosis. Early manifestation (<30 days) may be due to surgical technique. Nonanastomotic strictures are less common. Risk factors include HAS or HAT, chronic rejection, prolonged ischemia time, or recurrent disease in patients with underlying primary sclerosing cholangitis.

If a stricture cannot be crossed with a wire initially, an external biliary drain may be placed. Biliary inflammation often subsides after 1–2 weeks, at which point the stricture may be more amenable to being crossed. Once crossed, the stricture can be treated with balloon dilation and placement of a biliary drain across the stenosis (Fig E3). Use of cutting balloons for



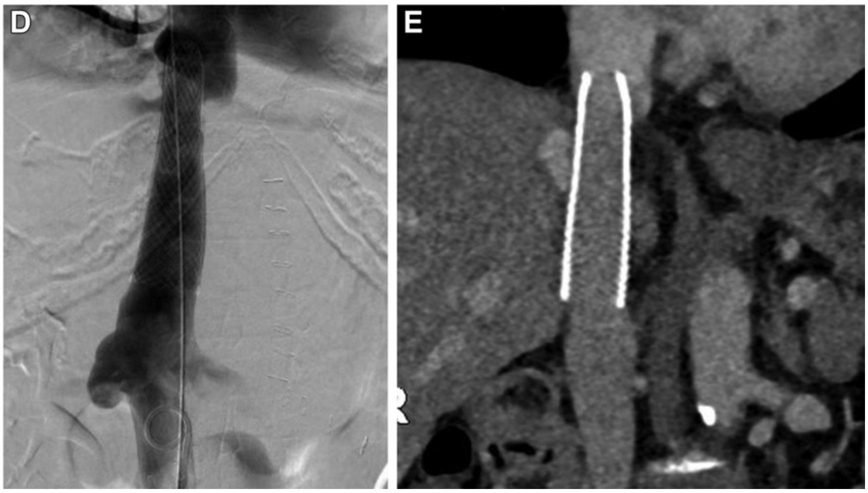


Figure 17. IVC thrombosis in a 64-year-old woman with upper abdominal pain 2 weeks after liver transplant. **(A)** Coronal contrast-enhanced CT image shows thrombus in the intrahepatic IVC with stenosis (arrow). **(B)** Venogram from a right femoral approach shows thrombus with minimal filling of the right atrium. After suction thrombectomy, there was improvement in clot burden with some residual filling defect. **(C)** Venogram shows the IVC stenosis (arrow.) A US-assisted thrombolysis catheter was placed, and an overnight thrombolytic infusion was performed. The next day, additional residual thrombus was removed using a rheolytic thrombectomy device, and the IVC stenosis was treated with stent placement. **(D)** Venogram shows the stent in the IVC. The patient's symptoms resolved, and liver function was preserved. **(E)** Coronal contrast-enhanced CT image 4 months later shows resolution of the thrombus and IVC stenosis with a patent stent and hepatic veins.

management of recalcitrant transplant biliary stenosis shows some promise in terms of technical success and safety (39).

Occasionally, a rendezvous procedure can be performed in conjunction with an endoscopist, depending on the anatomy and type of stricture, so that percutaneous drains can be converted into internal stents, which can then be managed endoscopically. If this is not possible, drain exchanges and drain upsizing along with balloon dilation are often required over many months for long-term successful treatment (Fig 18). One study that used such a protocol achieved patency of 84% at 1 year, 78% at 2 years, and 74% at 5 years. That study included liver transplant strictures and other benign biliary strictures (40).

A novel threefold protocol has shown some promise and offers the advantage of not requiring long-term external biliary drainage—a major advantage for patient comfort and quality of life. The stricture is crossed, and a 20-minute prolonged cholangioplasty is performed. This is repeated on days 3 and 5, and the catheter is removed. Patency rates were 56% at 1 year and 41% at 2 years (41). Strictures that fail interventional management may ultimately require surgical revision or retransplantation.

Ischemic Cholangiopathy

Ischemic cholangiopathy is characterized by diffuse injury to the biliary tree, which can result in multifocal nonanastomotic stenoses of the intrahepatic and extrahepatic ducts or more extreme cases with biliary debris and casts. Risk factors include HAS or HAT, immunologic rejection, donation-after-cardiac-death (DCD) liver, and prolonged ischemia time (42,43). Cholestasis can cause formation of gallstones, sludge, and debris, further obstructing the bile ducts, leading to cholangitis.

Ischemic cholangiopathy is not reversible, but there are management options that are similar to those for biliary stricture, namely percutaneous biliary drainage and balloon dilation of stenoses (Fig 19). Percutaneous access may also be used

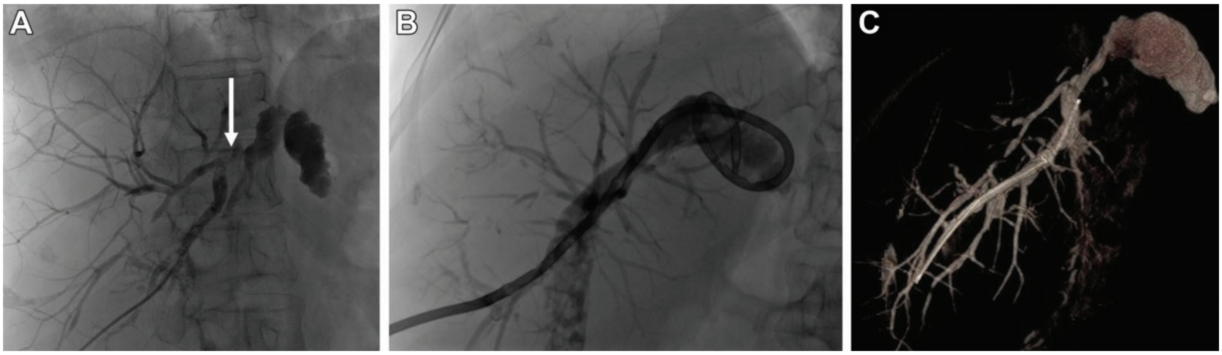


Figure 18. Biliary-enteric anastomotic stricture in a 70-year-old woman with recurrent episodes of cholangitis and elevated liver enzyme levels after living donor liver transplant 18 years earlier for primary sclerosing cholangitis. MR cholangiopancreatography (MRCP) showed dilated bile ducts and a hepaticojejunostomy anastomotic stenosis (not shown). **(A)** Percutaneous transhepatic cholangiogram shows mildly dilated bile ducts with a stenosis (arrow) at the biliary-enteric anastomosis. The stricture was crossed, and a 10-F internal-external biliary drainage catheter was placed. **(B)** Cholangiogram shows the placed biliary drainage catheter. After 1 month, the patient returned for cholangioplasty at the stricture. This was repeated monthly for 4 months. **(C)** Cone-beam CT three-dimensional reconstruction cholangiogram 6 months after initial drain placement shows resolution of the stenosis, with excellent flow into the jejunum.

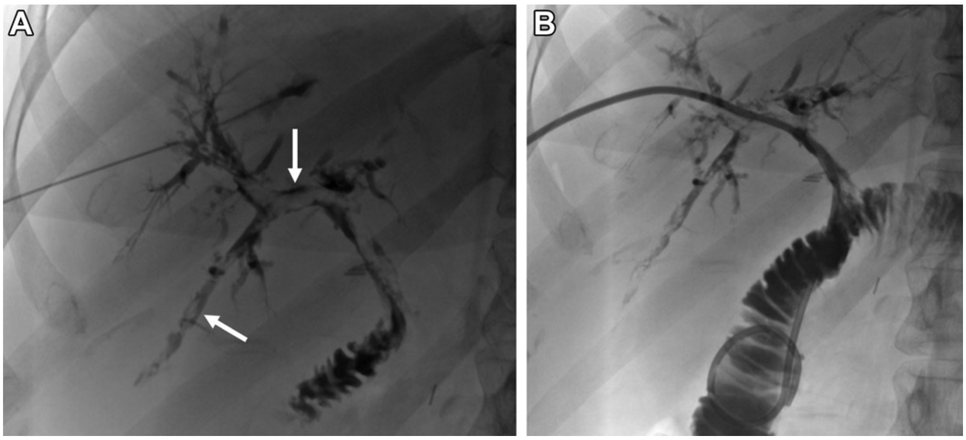


Figure 19. Ischemic cholangiopathy in a 56-year-old woman after deceased donor liver transplant with Roux-en-Y hepaticojejunostomy complicated by severe hepatic artery anastomotic stenosis. She presented with hyperbilirubinemia and concern for ischemic cholangiopathy. ERCP was unsuccessful owing to altered surgical anatomy. **(A)** Percutaneous transhepatic cholangiogram 2 months after the transplant shows irregularity and extensive debris (arrows) in the biliary system. Note the patency of the hepaticojejunostomy. An internal-external 8.5-F biliary drain was placed. **(B)** Cholangiogram shows the placed biliary drain. The patient was managed with internal-external drainage and routine exchanges every 4 weeks until retransplantation 7 months later.

for retrieval of obstructing debris or casts with a Fogarty balloon or basket or advanced retrieval techniques such as cholangioscopy (44). The goal of these interventions is to prevent obstruction and recurrent cholangitis.

Often, these patients may present with intrahepatic abscesses or bilomas that may require abscess drainage before biliary drainage. (Figs E4 and E5 show some management strategies.) Ultimately, these patients often require retransplantation for definitive treatment.

Bile Leak

Bile leaks occur in 2%–25% of posttransplant cases. Sources of leak include the biliary anastomosis, cystic duct, cut surface of the liver, and T-tube insertion site if present. Most leaks occur within 6 months after transplant. Early leaks (<30 days) may be related to surgical technical issues. Late leaks (>30 days) may be due to T-tube removal or underlying infection.

If symptomatic, bile leaks may manifest as abdominal pain, bile leakage from the incision, fever, peritonitis, or potentially sepsis. Diagnostic imaging is often pursued to identify discrete biloma collections that would require percutaneous drainage. Hepatobiliary iminodiacetic acid (HIDA) scintigraphy can also be performed to confirm the presence of a leak.

Management consists of percutaneous biloma drainage initially. When a leak is present, the outputs from the drainage catheter remain high and

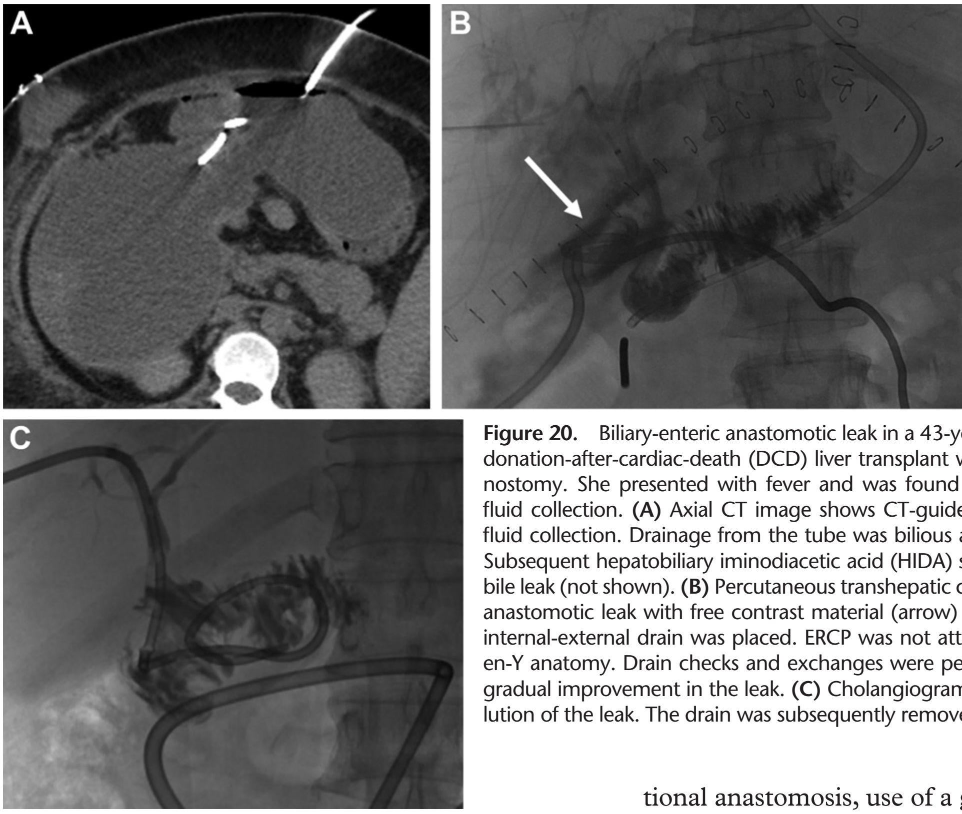


Figure 20. Biliary-enteric anastomotic leak in a 43-year-old woman 10 days after donation-after-cardiac-death (DCD) liver transplant with Roux-en-Y hepaticojejunostomy. She presented with fever and was found to have a large subhepatic fluid collection. **(A)** Axial CT image shows CT-guided drain placement into the fluid collection. Drainage from the tube was bilious and remained high for days. Subsequent hepatobiliary iminodiacetic acid (HIDA) scintigraphy demonstrated a bile leak (not shown). **(B)** Percutaneous transhepatic cholangiogram shows a large anastomotic leak with free contrast material (arrow) in the subhepatic space. An internal-external drain was placed. ERCP was not attempted owing to the Roux-en-Y anatomy. Drain checks and exchanges were performed every 4 weeks with gradual improvement in the leak. **(C)** Cholangiogram 5 months later shows resolution of the leak. The drain was subsequently removed.

will be bilious. If the leak is not amenable to endoscopic stent placement, percutaneous transhepatic cholangiography (PTC) can be performed with placement of a percutaneous internal-external drain across the leak (Fig 20), allowing healing and resolution over the course of months. In complex leaks that do not respond to biliary drainage, covered stents can be placed. In some situations, embolization can be performed using liquid embolics, coils, or alcohol as a sclerosant (45,46).

Update on Liver Transplant Interventions

Transplenic Portal Vein Recanalization

Chronic obliterative PVT is a known complication of portal hypertension in patients with cirrhosis, with a prevalence in these patients varying from 5% to 26% (47). Symptoms may reflect sequelae of portal hypertension, including gastrointestinal bleeding, ascites, splenomegaly, and intestinal venous congestion. Some patients may be asymptomatic. PVT remains an absolute or relative contraindication to liver transplantation in some centers and is an independent risk factor for posttransplant morbidity and mortality (48).

The primary goal in management of chronic PVT is to establish recanalized portal flow and improve transplant candidacy, so that portal flow can be restored through conventional end-to-end portal anastomosis (47). By allowing a conventional anastomosis, use of a graft or conduit can be avoided, both of which are associated with increased complications (49,50). Secondary goals involve reducing progression of thrombus extension throughout the portal vein or into the SMV and relieving any symptoms related to portal hypertension.

Percutaneous recanalization of a chronically occluded portal vein in conjunction with TIPS placement can be used to recanalize the portal vein and increase transplant candidacy. The technical success rate ranges from 70% to 100%. In a case series of 61 patients with PVT, the technical success rate was 98%. Portal vein–TIPS patency was maintained in 92% of patients at a median follow-up of 19.2 months. Thirty-nine percent of the patients underwent liver transplant (49).

At the authors' institution, transsplenic access to a central intraparenchymal splenic vein branch is accomplished with US guidance. Review of preoperative imaging is critical to find a splenic vein branch that is contiguous to the main splenic vein with a relatively direct path. Access is obtained with a 21-gauge needle, with upsizing to a 6-F sheath. Portal venography is performed.

As described by Thornburg et al (51) and in the authors' experience, the thrombosed portal vein can often be identified as a late-filling relatively straight structure at the superior aspect of the portal vein confluence (52). This is typically accessed with a 0.035-inch Glidewire (Terumo) into a right portal vein branch, which is exchanged for a snare device. Using a transjugular approach, a TIPS needle is advanced from the right hepatic

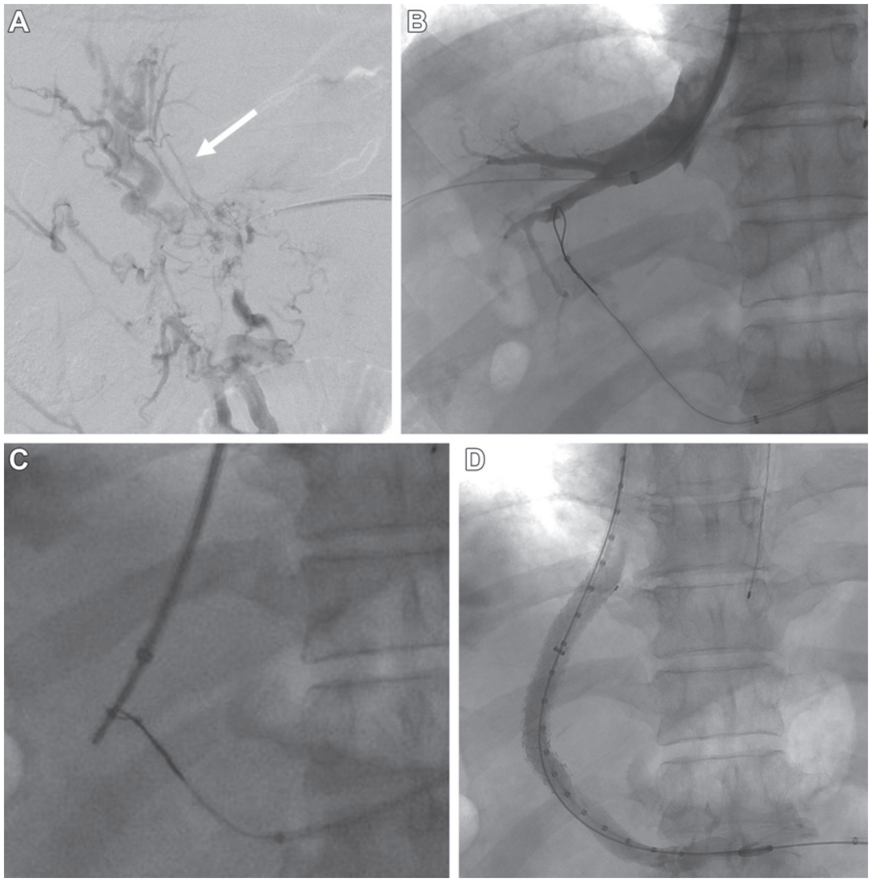


Figure 21. Transplenic portal vein recanalization in a 68-year-old man with cavernous transformation of the portal vein undergoing evaluation for liver transplant. **(A)** Transsplenic venogram shows filling of portal vein collaterals and a straighter and diminutive chronically thrombosed portal vein (arrow). A catheter was advanced into a right portal vein branch. **(B)** Venogram shows a sheath in the right hepatic vein and a gooseneck snare placed in a right portal vein branch. **(C)** A metal needle was advanced from the hepatic vein targeting the snare. A wire was advanced through the needle, snared, and pulled through the transsplenic sheath. Over the wire, a TIPS stent-graft is deployed and portal vein angioplasty is performed. **(D)** Completion venogram shows portal vein recanalization with a patent TIPS. TIPS portography 4 weeks later demonstrated portal vein patency (not shown). The patient subsequently underwent successful liver transplantation with end-to-end portal vein anastomosis.

vein through the snare, and a wire is advanced. The wire is then pulled through using the snare device so that it exits the splenic vein.

Over this wire, venoplasty and TIPS stent placement can occur. Placement is critical, minimizing stent coverage of the main portal vein, allowing remodeling to occur, and improving patency for a conventional end-to-end anastomosis (Fig 21). Embolization of the splenic tract is the critical last step, usually performed with coils, plugs, or Gelfoam (absorbable gelatin sponge; Pfizer).

Spontaneous Splenorenal Shunt

Approximately 20%–30% of liver transplant candidates have a spontaneous splenorenal shunt (SSRS) (53). Many SSRSs will resolve on their own after liver transplant as the portal pressure is reduced. However, large SSRSs (>10 mm in diameter) may persist in siphoning blood flow from the portal vein (54,55). This may result in portal hypoperfusion, leading to graft dysfunction, PVT, or encephalopathy.

Whether the presence of an SSRS has a negative impact on the liver transplant graft is controversial, with some studies showing that its presence has little impact on graft survival (56). Some centers will address shunts at the time of liver transplant. Intraoperative techniques include real-time assessment of portal flow with compression of the SSRS.

If portal flow is increased, the SSRS can be ligated. Alternatively, the left renal vein can be ligated to decrease shunting through the SSRS. Postoperatively, if the patient has a large SSRS with evidence of portal hypoperfusion, embolization of the shunt can be performed to increase portal flow. Data are limited with regard to results after embolization of an SSRS in the setting of liver transplant (Fig E6).

Splenic Artery Steal Syndrome

Also known as nonocclusive hepatic artery hypoperfusion syndrome, splenic artery steal syndrome is an underrecognized cause of liver dysfunction after liver transplant, as its manifestation mimics other causes of graft failure. It can be seen in up to 4% of transplant patients (32). The cause of hepatic dysfunction is controversial. Initially thought to be due to preferential flow to an enlarged splenic artery "stealing" from the transplant hepatic artery, it is now theorized to be due to portal vein hyperperfusion.

When portal flow is increased, arterial vasoconstriction occurs via the hepatic artery buffer response, leading to increased hepatic resistance and poor arterial flow (57). The combination of graft dysfunction, poor hepatic artery perfusion without structural obstruction, and increased flow in the portal vein is diagnostic for splenic artery

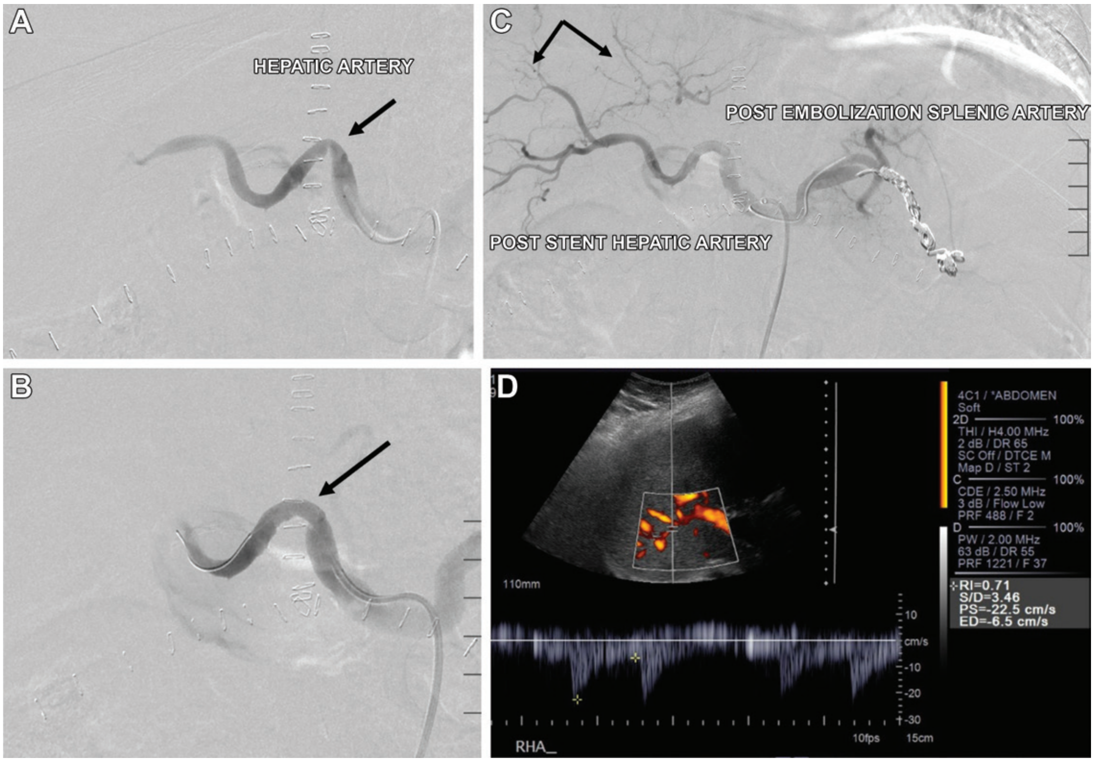


Figure 22. Splenic artery steal syndrome 3 months after liver transplant in a 46-year-old man with graft dysfunction. There was difficulty identifying arterial flow at US. **(A)** Hepatic angiogram shows HAS (arrow), which was thought to be the cause of poor arterial inflow. This was treated with stent placement. **(B)** Angiogram shows that, despite stent placement (arrow), there is poor opacification of intrahepatic branches with to-and-fro contrast material flow and shunting toward the splenic artery. Coil embolization of the splenic artery was performed. **(C)** Postembolization angiogram shows improved opacification of the distal hepatic artery branches (arrows). **(D)** Postprocedure Doppler US image the next day shows normal arterial waveforms.

steal syndrome after transplantation (58). Often, the entity is identified in the immediate weeks after transplantation, as greater than 80% of patients with this entity are diagnosed within the first 60 days of transplantation (57).

Initial evaluation is with Doppler US, which may show nonspecific increased resistive indexes in the hepatic artery. In some cases, elevated portal flow velocities have also been described. However, this is not consistently reported. The standard of reference for diagnosis remains angiography. At angiography, a large splenic artery is seen. The hepatic artery is usually characterized by sluggish flow without structural obstruction.

Endovascular embolization of the proximal to mid splenic artery can be performed so that collateral supply to the spleen can be preserved, thereby decreasing the risk of splenic infarction. Embolization is typically performed with coils or a vascular plug (32) (Fig 22).

TIPS after Liver Transplant

Development of ascites after liver transplant is often due to vascular issues such as outflow obstruction or increased portal vein flow. However, cirrhosis can develop in the transplant liver owing to recurrence of viral hepatitis, chronic rejection, or other factors. In such patients, TIPS for portal decompression can be considered. One study showed that approximately 2% of all liver transplant patients may develop portal hypertension and require a TIPS (59).

The clinical effectiveness of TIPS in the liver transplant patient is good, but not as good as the effectiveness in nontransplant patients. Therefore, patient selection is critical. A predictor of patient survival after TIPS is the model for end-stage liver disease (MELD) score. Patients with a MELD score of less than 15 have a better prognosis than patients with a MELD score greater than 15 (60).

Technically, there may be some difference in the orientation of the hepatic vein and IVC. Particularly in a patient with cavocavostomy, adjunct techniques such as intravascular US guidance, transhepatic hepatic vein access, or transhepatic portal vein access with the “gun-sight” method (61) can be useful. However, the technical success rate of TIPS creation in transplant patients is similar to that in nontransplant patients (59).

Conclusion

Interventional radiology plays a critical role in management of liver transplant patients. These cases can be complex, whereby understanding the

Table 2: Lessons Learned from Liver Transplant Interventions

Read the operative note and be aware of transplant anatomy before starting the case. Variations exist in the hepatic artery, portal vein, hepatic outflow, and biliary system. Work closely with your transplant surgeons and hepatology teams. Communicate often.
For HAS cases, CT angiography with 3D reconstruction can be invaluable for procedure planning. This helps plan the approach (radial or femoral) and demonstrates the best obliquity for imaging the HAS on angiograms.
HAS due to arterial kinking may not respond favorably to endovascular intervention. Consideration should be given to surgical revision.
A 0.014-inch (or less commonly 0.018-inch)–compatible system is useful for HAS cases. Guide catheters may be useful for stability in the celiac or common hepatic artery. Stiff wires can cause pseudostenosis when there is straightening of often tortuous arteries. Judicious use of intra-arterial nitroglycerin and intravenous heparin is recommended when treating HAS with balloons or stent placement.
Use of general anesthesia for initial PTC and biliary drain placement is useful, particularly if the intrahepatic biliary ducts are nondilated.
Ischemic cholangiopathy cases are some of the most challenging to manage. These patients require imaging surveillance and may benefit from more frequent biliary tube exchange (every 4 weeks instead of every 8–10 weeks).
Cone-beam CT can be a useful tool for biliary interventions. It can help identify which segments of the liver are being adequately drained or undrained. Also, it may help identify a better obliquity for identifying or crossing a stenosis.
Fluid collections in the perioperative period will often resolve on their own.
Consider transsplenic access for portal vein interventions. This has the advantage of avoiding a transhepatic puncture.
With side-to-side cavocavostomy anatomy, as compared with the bicaval or piggyback technique, gaining access to the hepatic veins for a TIPS or transjugular liver biopsy may be challenging, as cannulation requires a cranial course before the hepatic vein orifice. This can be facilitated by using a reverse-curve catheter or cobra-shaped catheter. Alternatively, a transfemoral venous approach can be used.

Note.—PTC = percutaneous transhepatic cholangiography, 3D = three-dimensional.

surgical anatomy, indications for intervention, technical aspects, and expected results is imperative. Lessons learned and tips gained from our experience are listed in Table 2. Increased comprehension of liver transplant physiology and development of new techniques continue to evolve and mature, improving outcomes for liver graft recipients.

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